

1. What is NLP?

Answer:

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, process, and generate human language.

2. What are the applications of NLP?

Answer:

- Chatbots
 - Machine Translation
 - Sentiment Analysis
 - Text Summarization
 - Speech Recognition
 - Spam Detection
 - Search Engines
-

3. What is Text Preprocessing?

Answer:

Text preprocessing is the process of cleaning and converting raw text into a format suitable for analysis.

Steps:

- Tokenization
 - Stopword Removal
 - Stemming
 - Lemmatization
 - Lowercasing
-

TOKENIZATION

4. What is Tokenization?

Answer:

Tokenization is the process of breaking text into smaller units called tokens.

Example:

Text: I am Omkar

Tokens: ["I", "am", "Omkar"]

5. Why is Tokenization important?

Answer:

It converts text into manageable pieces for NLP algorithms.

6. Which library did you use?

Answer:

NLTK using:

`word_tokenize()`

`sent_tokenize()`

Code Explanation

```
from nltk.tokenize import word_tokenize
```

```
text = "Hello Everyone"
```

```
tokens = word_tokenize(text)
```

```
print(tokens)
```

Explanation:

- Import tokenizer
- Store text
- Split into words
- Print tokens

STOPWORD REMOVAL

7. What are Stopwords?

Answer:

Frequently occurring words that carry little meaning.

Examples:

is, am, are, the, an, a

8. Why remove stopwords?

Answer:

To reduce noise and improve processing efficiency.

9. Which NLTK module is used?

Answer:

```
from nltk.corpus import stopwords
```

STEMMING

10. What is Stemming?

Answer:

Reducing a word to its root form by removing suffixes.

Example:

Playing → Play

Running → Run

11. What stemmers did you use?

Answer:

1. Porter Stemmer
2. Lancaster Stemmer

12. Difference between Porter and Lancaster?

Porter	Lancaster
Less aggressive	More aggressive
More accurate	Faster

Example:
Living
Porter → live
Lancaster → liv

LEMMATIZATION

13. What is Lemmatization?

Answer:

Converts words into their dictionary form (lemma).

Example:

Running → Run
Children → Child

14. Difference between Stemming and Lemmatization?

Stemming	Lemmatization
Rule-based	Dictionary-based
Faster	More accurate
May create invalid words	Produces valid words

15. Which libraries were used?

Answer:

- NLTK WordNetLemmatizer
 - spaCy
-

N-GRAM

16. What is an N-Gram?

Answer:

A sequence of N consecutive words.

Example:

I am Omkar

Unigram

I

am

Omkar

Bigram

I am

am Omkar

Trigram

I am Omkar

17. Why are N-Grams used?

Answer:

- Next word prediction
 - Language modeling
 - Text generation
-

18. What is Bigram Probability?

Formula:

$$P(w_2|w_1) = \text{Count}(w_1, w_2) / \text{Count}(w_1)$$

POS TAGGING

19. What is POS Tagging?

Answer:

Assigning grammatical tags to words.

Example:

I → PRP

am → VBP

student → NN

20. Common POS Tags

Tag Meaning

NN Noun

VB Verb

JJ Adjective

RB Adverb

PRP Pronoun

21. Which library did you use?

Answer:

nlk.pos_tag()

and spaCy.

NER

22. What is NER?

Answer:

Named Entity Recognition identifies important entities in text.

Example:

Omkar lives in Mumbai

Output:

Omkar → PERSON

Mumbai → LOCATION

23. Common NER Labels

Label	Meaning
-------	---------

PERSON	Person Name
--------	-------------

ORG	Organization
-----	--------------

GPE	Location
-----	----------

DATE	Date
------	------

24. What is Custom NER?

Answer:

Creating your own entity labels and training a model on custom data.

Example:

SKILL

COLLEGE

PRODUCT

TEXT REPRESENTATION

25. What is Text Representation?

Answer:

Converting text into numerical vectors so machines can understand it.

Methods:

1. BoW
 2. TF-IDF
 3. Word Embeddings
-

BAG OF WORDS

26. What is Bag of Words (BoW)?

Answer:

Represents text based on word frequencies.

Example:

I love NLP

I love AI

Vocabulary:

I, love, NLP, AI

Vector:

[2,2,1,1]

27. Limitation of BoW?

Answer:

- Ignores word order
- Produces sparse vectors

TF-IDF

28. What is TF-IDF?

Formula:

$$\text{TF-IDF} = \text{TF} \times \text{IDF}$$

29. Full Form?

Answer:

Term Frequency – Inverse Document Frequency

30. Why use TF-IDF?

Answer:

Gives more importance to meaningful words and less to common words.

WORD EMBEDDINGS

31. What are Word Embeddings?

Answer:

Dense vector representations of words capturing semantic meaning.

32. Examples?

Answer:

- Word2Vec
 - GloVe
 - FastText
-

33. Advantage over BoW?

Answer:

Captures semantic relationships.

Example:

King - Man + Woman $\approx$ Queen

TEXT CLASSIFICATION

34. What is Text Classification?

Answer:

Assigning categories to text.

Examples:

- Spam / Ham
 - Positive / Negative
 - Sports / Politics
-

35. Which algorithm did you use?

Answer:

Multinomial Naive Bayes

36. Why Naive Bayes?

Answer:

- Fast
- Simple
- Works well for text classification

SENTIMENT ANALYSIS

37. What is Sentiment Analysis?

Answer:

Determining emotional tone of text.

Classes:

- Positive
- Negative
- Neutral

Example

I love this phone.

Positive

I hate this phone.

Negative

TEXT SUMMARIZATION

38. What is Text Summarization?

Answer:

Generating a shorter version of a text while preserving key information.

39. Types of Summarization?

Extractive

Selects important sentences.

Abstractive

Generates new sentences.

MOST IMPORTANT CODE VIVA QUESTIONS

Why do we use `nlk.download('punkt')`?

To download tokenizer models.

Why use `CountVectorizer`?

To convert text into numerical vectors.

Why use `TfidfVectorizer`?

To generate TF-IDF features.

What is fit_transform()?

- fit() learns vocabulary
- transform() converts text
- fit_transform() does both

Difference between fit() and transform()?

fit()	transform()
Learns data	Converts data

What is a Corpus?

A collection of documents/texts.

What is Vocabulary?

All unique words present in a corpus.

What is Cosine Similarity?

Formula:

$$A.B / |A||B|$$

Used to measure similarity between documents.

EXAMINER'S FAVORITE RAPID FIRE QUESTIONS

1. NLP Full Form?
 - Natural Language Processing
2. NLTK Full Form?
 - Natural Language Toolkit
3. spaCy is?
 - NLP Library
4. Word2Vec developed by?
 - Google
5. TF-IDF Full Form?
 - Term Frequency Inverse Document Frequency
6. POS Full Form?
 - Part of Speech
7. NER Full Form?
 - Named Entity Recognition
8. BoW Full Form?
 - Bag of Words
9. What is a Token?
 - Smallest text unit
10. What is Lemma?
 - Dictionary form of a word

Q1. Tokenization Program Viva Questions

What is word_tokenize()?

It splits a sentence into individual words (tokens).

What is sent_tokenize()?

It splits a paragraph into sentences.

Why do we use nltk.download('punkt')?

To download the tokenizer model required by NLTK.

Difference between word tokenization and sentence tokenization?

- Word Tokenization → breaks text into words.
- Sentence Tokenization → breaks text into sentences.

Input

```
text = "Hello Everyone My name is Omkar."
```

Output of word_tokenize()

```
['Hello', 'Everyone', 'My', 'name', 'is', 'Omkar', '.']
```

Output of sent_tokenize()

```
['Hello Everyone My name is Omkar.']
```

Why is tokenization the first step in NLP?

Because NLP algorithms work on tokens rather than raw text.

Q2. Stop Word Removal Viva Questions

What are stop words?

Frequently occurring words with little meaning.

Examples:

is, am, are, the, a, an

Which library contains stop words?

```
from nltk.corpus import stopwords
```

What does this line do?

```
stop_words = set(stopwords.words('english'))
```

Creates a collection of English stop words.

Why use set()?

Because searching in a set is faster than in a list.

Explain this code:

```
if word.lower() not in stop_words:
```

Converts the word to lowercase and checks whether it is a stop word.

Why remove stop words?

- Reduce noise
- Improve accuracy
- Reduce processing time

Example

Input:

I am living in Mumbai

After stopword removal:

living Mumbai

Q3. Stemming Viva Questions

What is stemming?

Removing prefixes/suffixes to get the root form.

Which stemmers are used?

PorterStemmer()

LancasterStemmer()

Why use stemming?

To reduce different forms of a word into one form.

Example:

Playing

Played

Plays

becomes

Play

Explain this line:

porter.stem(word)

Returns the stemmed version of the word using Porter algorithm.

Difference between Porter and Lancaster?

Porter	Lancaster
--------	-----------

Less aggressive	More aggressive
-----------------	-----------------

More accurate	Faster
---------------	--------

Example

running

Porter:

run

Lancaster:

run

Limitation of Stemming?

Sometimes produces invalid words.

Example:

University → univers

Q4. Lemmatization Viva Questions

What is Lemmatization?

Converting a word to its dictionary form.

Difference between stemming and lemmatization?

Stemming	Lemmatization
----------	---------------

Cuts words	Uses dictionary
------------	-----------------

Faster	More accurate
--------	---------------

May create invalid words	Produces valid words
--------------------------	----------------------

What is a lemma?

Dictionary form of a word.

Example:

children → child

running → run

What does this line do?

```
lemmatizer.lemmatize(w)
```

Returns the root word.

Why specify POS?

```
lemmatizer.lemmatize(w, pos="v")
```

To tell WordNet that the word is a verb.

What are common POS values?**POS Meaning**

n noun

v verb

a adjective

r adverb

Why is spaCy lemmatization better?

Because it uses statistical models and context.

Example:

running

Output:

run

Q5. N-Gram Viva Questions**What is an N-Gram?**

A sequence of N consecutive words.

What is Unigram?

Single word.

Example:

I

am

Omkar

What is Bigram?

Two consecutive words.

Example:

I am

am Omkar

What is Trigram?

Three consecutive words.

Example:

I am Omkar

What does this line do?

ngrams(tokens, 2)

Generates bigrams.

Why use N-Grams?

- Next word prediction
- Auto-complete
- Language modeling

What is Counter() used for?

Counter(bigrams)

Counts frequencies of bigrams.

Explain this formula:

bigram_freq[(w1,w2)] / unigram_freq[w1]

Calculates Bigram Probability:

$$P(w_2|w_1) = \frac{\text{Count}(w_1, w_2)}{\text{Count}(w_1)}$$

What is the purpose of predict_next_word()?

Predicts the most probable next word based on bigram frequency.

Example

Input:

I am Omkar

I am Student

If input word is:

I

Prediction:

am

Very Important "Why" Questions**Why do we import NLTK?**

To perform NLP tasks like tokenization, stemming, POS tagging, etc.

Why do we use lowercase conversion?

To treat "Hello" and "hello" as the same word.

Why is preprocessing necessary?

To clean text before applying machine learning.

Why is tokenization important before stopword removal?

Because stopwords can only be removed after text is split into tokens.

Why is lemmatization preferred over stemming?

Because it produces meaningful dictionary words.

One-Line Viva Answers**Tokenization?**

→ Splitting text into tokens.

Stopword?

→ Common words with little meaning.

Stemming?

→ Reducing words to root form by cutting suffixes.

Lemmatization?

→ Converting words to dictionary form.

N-Gram?

→ Sequence of N consecutive words.

Bigram?

→ Two consecutive words.

Trigram?

→ Three consecutive words.

Porter Stemmer?

→ Popular stemming algorithm.

WordNet?

→ Lexical database used in lemmatization.

spaCy?

→ Industrial-strength NLP library.

POS Tagging (NLTK)

What is POS Tagging?

POS (Part of Speech) Tagging is the process of assigning grammatical categories to words such as noun, verb, adjective, etc.

Explain this code:

```
from nltk import pos_tag  
pos_tags = nltk.pos_tag(tokenize)
```

Answer:

- pos_tag() assigns POS tags to each token.
- It returns tuples containing (word, tag).

Example:

I am Omkar

Output:

```
[('I','PRP'),('am','VBP'),('Omkar','NNP')]
```

Why do we download?

```
nltk.download('averaged_perceptron_tagger_eng')
```

Answer:

It downloads the pretrained POS tagging model used by NLTK.

Common POS Tags

Tag Meaning

NN Noun

NNP Proper Noun

VB Verb

VBP Verb Present

JJ Adjective

RB Adverb

PRP Pronoun

Difference between NN and NNP?

NN = Common noun

Example:

student

city

NNP = Proper noun

Example:

Omkar

Mumbai

India

POS Tagging (spaCy)

Explain this code:

for token in doc:

```
    print(token.text, token.pos_, token.tag_)
```

Answer:

It prints:

- Word
- Universal POS tag (pos_)
- Detailed POS tag (tag_)

Example:

Omkar NOUN NN

Difference between pos_ and tag_ ?

pos_	tag_
General category	Detailed category
NOUN	NN
VERB	VBP

Why use spaCy instead of NLTK?

Answer:

- Faster
- More accurate
- Industrial NLP library
- Supports NER, parsing, embeddings

Named Entity Recognition (NER)

What is NER?

NER identifies named entities in text.

Example:

Omkar lives in Mumbai

Output:

Omkar → PERSON

Mumbai → GPE

What does this line do?

```
doc = nlp(user_sentence)
```

Answer:

Processes text using spaCy NLP pipeline.

Explain:

token.ent_type_

Answer:

Returns entity type.

Example:

PERSON

ORG

GPE

DATE

Explain:

token.ent_iob_

Answer:

Returns token position inside entity.

Possible values**Value Meaning**

B Beginning

I Inside

O Outside

What is GPE?**Answer:**

Geo-Political Entity

Examples:

India

Mumbai

USA

Difference between POS Tagging and NER?**POS**

Finds grammar

Noun, Verb

Language structure

NER

Finds entities

Person, City

Information extraction

spaCy Tokenization**Why use?**

nlp = spacy.blank("en")

Answer:

Creates a blank English language model without pretrained components.

Explain:

doc = nlp("Hello World")

Answer:

Converts text into a spaCy document object.

What is Doc object?**Answer:**

Container holding tokens and linguistic information.

What is Token object?**Answer:**

Represents one word or punctuation mark.

Custom NER Training**Why use?**

```
nlp = spacy.blank("en")
```

Answer:

To create a custom model from scratch.

Explain:

```
ner = nlp.add_pipe("ner")
```

Answer:

Adds Named Entity Recognition component to pipeline.

Explain:

```
ner.add_label("ORG")
```

Answer:

Registers a custom entity label.

Example:

OpenAI → ORG

What is TRAIN_DATA?

```
TRAIN_DATA = [  
    ("OpenAI builds AI models",  
     {"entities":[(0,6,"ORG")]})  
]
```

Answer:

Training examples with entity annotations.

Explain annotation format

(0,6,"ORG")

Meaning:

Start Position = 0

End Position = 6

Entity Type = ORG

Explain:

`Example.from_dict()`

Answer:

Converts training annotations into spaCy Example object.

Explain:

`nlp.update()`

Answer:

Updates model weights during training.

What are Losses?

`losses={}`

Answer:

Shows model error during training.

Viva Answer

"Lower loss means the model is learning better."

Why run multiple epochs?

`for epoch in range(20)`

Answer:

To improve model learning through repeated training.

Explain:

`nlp.to_disk("custom_ner_model")`

Answer:

Saves trained model permanently.

Custom PERSON-CITY NER Questions**Why add labels?**

`ner.add_label("PERSON")`

`ner.add_label("CITY")`

Answer:

To train the model to identify persons and cities.

Explain output

John -> PERSON

Mumbai -> CITY

Answer:

Model correctly identifies named entities based on training data.

Most Expected Viva Questions**What is spaCy?**

Open-source NLP library.

What is en_core_web_sm?

Pretrained English language model.

What is a pipeline?

Sequence of NLP processing steps.

What is a Named Entity?

Real-world object like person, city, organization.

What is ORG?

Organization.

What is PERSON?

Human name.

What is CITY/GPE?

Location or city.

What is Example object in spaCy?

Training example used for model learning.

Why do losses decrease during training?

Because model predictions become more accurate.

What is the difference between pretrained NER and custom NER?

Pretrained NER

- Already trained
- Generic entities

Custom NER

- Trained by us
- Domain-specific entities

1. Creating and Comparing Different Text Representations

What is Text Representation?

Answer:

Text representation is the process of converting text into numerical vectors so that machine learning algorithms can process it.

Examples:

- Bag of Words (BoW)
 - TF-IDF
 - Word Embeddings
-

Bag of Words (BoW)

What is Bag of Words?

Answer:

BoW represents text using word frequencies without considering word order.

Example:

Sentence 1:

I love NLP

Sentence 2:

I love AI

Vocabulary:

[I, love, NLP, AI]

Vectors:

[1,1,1,0]

[1,1,0,1]

Explain this code:

```
from sklearn.feature_extraction.text import CountVectorizer
```

Answer:

CountVectorizer converts text into BoW vectors.

Explain:

```
count_vectorizer = CountVectorizer(max_features=1000)
```

Answer:

Creates a BoW vectorizer and keeps maximum 1000 important words.

Explain:

```
bow_matrix = count_vectorizer.fit_transform(data)
```

Answer:

- fit() learns vocabulary
 - transform() converts text to vectors
 - fit_transform() performs both operations
-

What is Vocabulary?

Answer:

All unique words in the corpus.

Example:

I love NLP

I love AI

Vocabulary:

I, love, NLP, AI

Limitation of BoW?**Answer:**

- Ignores word order
 - Ignores context
 - Creates sparse vectors
-

Cosine Similarity**What is Cosine Similarity?****Answer:**

Measures similarity between two vectors.

Formula:

$$\cos \theta = \frac{A \cdot B}{|A||B|}$$

Range:

0 → Completely Different

1 → Exactly Same

Why use Cosine Similarity?**Answer:**

To compare document similarity.

Example:

- Search engines
 - Recommendation systems
-

Explain:

`cosine_similarity(doc_0_bow, doc_1_bow)`

Answer:

Calculates similarity between two documents using BoW vectors.

TF-IDF**Full Form of TF-IDF?****Answer:**

Term Frequency – Inverse Document Frequency

What is TF?

Formula:

TF = Frequency of word in document

What is IDF?

Formula:

$$\text{IDF} = \log(\text{Total Documents} / \text{Documents containing word})$$

Why use TF-IDF?

Answer:

To give higher importance to meaningful words and lower importance to common words.

Example:

the, is, am

receive low scores.

Explain this code:

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

Answer:

Imports TF-IDF vectorizer.

Explain:

```
tfidf_vectorizer = TfidfVectorizer(max_features=1000)
```

Answer:

Creates TF-IDF vectorizer with maximum 1000 features.

Explain:

```
tfidf_matrix = tfidf_vectorizer.fit_transform(data)
```

Answer:

Converts text documents into TF-IDF vectors.

Difference between BoW and TF-IDF?

BoW

Counts words

Common words dominate

Simpler

TF-IDF

Weighs words

Common words get lower weight

More informative

Word Embeddings

What are Word Embeddings?

Answer:

Dense numerical vectors that capture semantic meaning of words.

Example:

King $\approx$ Queen

Man $\approx$ Woman

Why are Word Embeddings better than BoW?

Answer:

They capture semantic relationships between words.

Example:

King and Queen

will have similar vectors.

Which embedding model is used?

Answer:

glove-wiki-gigaword-50

(Pretrained GloVe model)

Explain:

```
import gensim.downloader as api
```

Answer:

Used to download pretrained embedding models.

Explain:

```
word_vectors = api.load("glove-wiki-gigaword-50")
```

Answer:

Loads pretrained GloVe embeddings having 50 dimensions.

What is vector dimension?

Answer:

Number of values representing a word.

Example:

50 dimensions

100 dimensions

300 dimensions

Explain:

```
word_vectors["computer"]
```

Answer:

Returns embedding vector for the word "computer".

Explain:

```
word_vectors.most_similar("king")
```

Answer:

Returns words having similar meanings.

Example:

queen

prince

monarch

Explain:

```
word_vectors.most_similar(  
positive=["woman","king"],  
negative=["man"]  
)
```

Answer:

Performs word analogy.

Result:

queen

What is Word Analogy?

Answer:

King - Man + Woman = Queen

This demonstrates semantic understanding by embeddings.

Sentence Embeddings

Why create sentence vectors?

Answer:

Machine learning models require fixed-length numerical vectors.

Explain:

```
def get_sentence_vector(sentence, model):
```

Answer:

Creates one vector representing the whole sentence.

Explain:

```
words = sentence.lower().split()
```

Answer:

Converts sentence to lowercase and splits into words.

Explain:

```
return np.mean(vecs, axis=0)
```

Answer:

Computes average of all word vectors to obtain sentence embedding.

Why use average vectors?

Answer:

Simple and effective method to represent entire sentences.

Most Important Viva Questions

Difference between BoW and Word Embeddings?

BoW	Word Embeddings
------------	------------------------

Frequency-based	Meaning-based
-----------------	---------------

Sparse vectors	Dense vectors
----------------	---------------

No semantics	Captures semantics
--------------	--------------------

Difference between TF-IDF and Word Embeddings?

TF-IDF	Embeddings
---------------	-------------------

Statistical	Semantic
-------------	----------

Word importance	Word meaning
-----------------	--------------

Sparse vectors	Dense vectors
----------------	---------------

Why is GloVe called Global Vectors?

Answer:

Because it learns word relationships using global word co-occurrence statistics.

What is a Dense Vector?

Answer:

A vector where most values are non-zero.

What is a Sparse Vector?

Answer:

A vector containing mostly zeros.

Examiner's Favorite Rapid-Fire Questions

1. BoW Full Form?
 - Bag of Words
2. TF-IDF Full Form?
 - Term Frequency Inverse Document Frequency
3. What is CountVectorizer?
 - Creates BoW vectors
4. What is TfidfVectorizer?
 - Creates TF-IDF vectors
5. What is Cosine Similarity?
 - Measures similarity between vectors
6. What is GloVe?
 - Word embedding model
7. What is Word2Vec?
 - Neural-network-based word embedding model
8. What is an Embedding?
 - Numerical representation of text
9. What is a Corpus?

- Collection of documents

10. What is Vocabulary?

- Set of unique words in a corpus

Q1. Text Classification

What is Text Classification?

Answer:

Text Classification is the process of assigning predefined categories to text documents.

Examples:

- Spam / Not Spam
 - Positive / Negative
 - Sports / Politics
-

Which algorithm is used in your code?

MultinomialNB()

Answer:

Multinomial Naive Bayes classifier.

Why is Naive Bayes used for NLP?

Answer:

- Fast
 - Simple
 - Works well with text data
 - Requires less training data
-

Explain:

```
X = df["Text"]
```

```
y = df["Label"]
```

Answer:

- $X \rightarrow$ Input features (text)
 - $y \rightarrow$ Target labels (positive/negative)
-

Explain:

```
train_test_split(X,y,test_size=0.2)
```

Answer:

Splits dataset into training and testing data.

Example

80% $\rightarrow$ Training

20% $\rightarrow$ Testing

Why do we split data?

Answer:

To evaluate model performance on unseen data.

Explain:

```
vectorizer = CountVectorizer()
```

Answer:

Converts text into numerical vectors using Bag of Words.

Explain:

```
X_train_vectorized = vectorizer.fit_transform(X_train)
```

Answer:

Creates feature vectors from training text.

Explain:

```
classifier.fit(X_train_vectorized,y_train)
```

Answer:

Trains the Naive Bayes model.

Explain:

```
y_pred = classifier.predict(X_test_vectorized)
```

Answer:

Predicts class labels for test data.

Explain:

```
classification_report(y_test,y_pred)
```

Answer:

Displays:

- Precision
 - Recall
 - F1-Score
 - Accuracy
-

Classification Metrics**What is Accuracy?**

Formula:

$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$

What is Precision?

Formula:

$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

Viva Answer

"Out of predicted positives, how many were actually positive."

What is Recall?

Formula:

$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Viva Answer

"Out of actual positives, how many were correctly identified."

What is F1 Score?

Formula:

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Difference between Precision and Recall?

Precision

Recall

False positives matter False negatives matter

Most Asked Viva Question

Why CountVectorizer before Naive Bayes?

Answer:

Machine learning algorithms cannot understand text directly. CountVectorizer converts text into numerical form.

Q2. Sentiment Analysis

What is Sentiment Analysis?

Answer:

Process of determining emotional tone of text.

Classes:

- Positive
 - Negative
 - Neutral
-

Example

Input:

I love this phone.

Output:

Positive

Input:

I hate this phone.

Output:

Negative

Which algorithm is used?

MultinomialNB()

Naive Bayes Classifier.

Explain:

```
df_sentiment = pd.read_csv()
```

Answer:

Loads sentiment dataset.

Explain:

`X_sentiment = df_sentiment["Text"]`

Answer:

Stores review text.

Explain:

`y_sentiment = df_sentiment["Label"]`

Answer:

Stores sentiment labels.

Explain:

`stratify=y_sentiment`

Answer:

Maintains equal class distribution in train and test data.

Why use CountVectorizer?**Answer:**

To convert text into numerical vectors.

Explain:

`classifier_sentiment.fit()`

Answer:

Trains sentiment model.

Explain:

`classifier_sentiment.predict()`

Answer:

Predicts sentiment.

Why evaluate sentiment model?**Answer:**

To measure prediction quality using:

- Accuracy
 - Precision
 - Recall
 - F1-score
-

Q3. Text Summarization**What is Text Summarization?****Answer:**

Process of generating a shorter version of a document while preserving important information.

Types of Summarization

1. Extractive Summarization

Answer:

Selects important sentences from original text.

Example:

Original:

NLP is a branch of AI.

It is used in chatbots.

It is also used in translation.

Summary:

NLP is a branch of AI.

It is used in chatbots.

2. Abstractive Summarization

Answer:

Generates new sentences similar to human writing.

Example:

Original:

Machine learning helps computers learn from data.

Summary:

Computers learn patterns from data.

Extractive Summarization Code Questions

Explain:

from collections import Counter

Answer:

Used for word frequency counting.

Explain:

stopwords = set(...)

Answer:

Stores common words to ignore.

Explain:

sentences = re.split(...)

Answer:

Splits text into sentences.

Explain:

freq = Counter(words)

Answer:

Counts frequency of each word.

Explain:

scores = {}

Answer:

Stores sentence importance scores.

Explain:

sorted(scores,key=scores.get,reverse=True)

Answer:

Sorts sentences by importance.

Explain:

top = sorted(...)[:num_sentences]

Answer:

Selects top scoring sentences.

Abstractive Summarization Questions

Which library is used?

transformers

Which model is used?

AutoModelForSeq2SeqLM

Explain:

AutoTokenizer.from_pretrained()

Answer:

Loads tokenizer.

Explain:

model.generate()

Answer:

Generates summary.

Explain parameters

num_beams=4

Beam Search width.

min_length=30

Minimum summary length.

max_length=120

Maximum summary length.

Explain:

`tokenizer.decode()`

Answer:

Converts tokens back into readable text.

Very Important Viva Questions**Difference between Text Classification and Sentiment Analysis?****Text Classification Sentiment Analysis**

General categories Emotion detection

Sports, News Positive, Negative

Difference between Extractive and Abstractive Summarization?**Extractive Abstractive**

Copies sentences Generates new sentences

Simpler More advanced

Rule based Deep Learning

Why is Naive Bayes popular in NLP?

- Fast
 - Simple
 - Handles large vocabulary
 - Works well with text data
-

Examiner's Rapid Fire Questions**What is NLP?**

Natural Language Processing

What is Classification?

Assigning labels to text.

What is Sentiment Analysis?

Detecting emotions from text.

What is Naive Bayes?

Probabilistic classifier.

What is Accuracy?

Correct predictions percentage.

What is Precision?

How many predicted positives are correct.

What is Recall?

How many actual positives are found.

What is F1 Score?

Balance of precision and recall.

What is Extractive Summarization?

Selecting important sentences.

What is Abstractive Summarization?

Generating new summary sentences.

What is CountVectorizer?

Converts text to BoW vectors.

What is Transformer?

Deep learning architecture used in modern NLP models.

If these four assignments come in viva, focus especially on:

1. Tokenization, Stopwords, Stemming, Lemmatization
2. POS Tagging and NER
3. BoW, TF-IDF, Word2Vec/GloVe
4. Text Classification, Sentiment Analysis, Summarization